ROLL NO: | | | | | | | | | | |

(To fill the roll no is compulsory)

INVERTIS UNIVERSITY BAREILLY

EVEN SEMESTER EXAMINATION - 2024-25

B.Tech. (AI)

Semester: II

SUBJECT CODE: BCSAI205

DIGITAL ELECTRONICS

Time: 03:00 Hrs. **Maximum Marks: 50**

NOTE: THE QUESTION PAPER CONTAINS THREE SECTIONS. ATTEMPT ALL THE SECTIONS.

SECTION – A

NOTE: Attempt all parts of the following question. **(10 Marks)**

Q. 1. Answer the following questions: **(1 × 5 = 5 Marks)**

(i) Define digital electronics and mention its applications.

(ii) What is logic gate? Give an example.

(iii) What are universal gates? Name them.

(iv) Convert $(101101)_2$ into decimal and octal number systems.

(v) What is a flip-flop?

Q. 2. Answer the following questions: **(1 × 5 = 5 Marks)**

(i) Write the truth table for an XOR gate.

(ii) Explain De Morgan's Theorem.

(iii) Explain Synchronous circuit.

(iv) Find 2's complement of $(11011)_2$

(v) State the necessity of Multiplexer.

SECTION – B

NOTE:Attempt all questions. **(5×2×2 = 20 Marks)**

Q.3.(i)(a) Minimise the following expression using K-map

$$f(P, Q, R, S) = \Sigma m(0, 3, 4, 5, 9, 11, 15) + d(8, 10, 14)$$

OR

(b) Simplify the following expression in POS form:

$$AB\bar{D} + \bar{C}D + \bar{B}D + ABCD$$

(ii)(a) Explain J-K flip flop in detail with characteristic table and also draw excitation table for it.

OR

(b) What are shift registers? Write the difference between SISO and SIPO registers.

Q.4.(i)(a) Draw and explain 4-bit binary-adder subtractor in detail with example.

OR

(b) Explain Full Adder in detail also draw K-map and circuit diagram for full adder. Justify Full adder is a combination of two half adders.

(ii)(a) Design 4:1 multiplexer using 8:1 multiplexer. Write the difference between multiplexer and demultiplexer.

OR

(b) Explain Decoder. Draw 3:8 Decoder using 2:4 Decoder with circuit diagram and truth table.

SECTION – C

NOTE: Attempt all questions. **(10 × 2 = 20 Marks)**

Q.5.(a) Minimise the following function using Quine-Mc Cluskey method.

$$F(A,B,C,D) = \Sigma m(0, 1, 3, 7, 8, 9, 10, 11, 15)$$

OR

(b) Draw the circuit diagram of a master-slave JK flip-flop and explain its working during the high and low phases of the clock. Also, discuss how the master-slave configuration eliminates the race condition present in conventional JK flip-flops.

Q.6.(a) Draw and explain 3-bit Asynchronous down counter with diagram and logic circuit and also explain operation of 4:2 priority encoder.

OR

(b) Convert the number $(11001010)_2$ into BCD, octal, decimal and Hexadecimal.